ADAPTING COVER SONG IDENTIFICATION METHODS TO THE UNIQUE FEATURES OF INDIAN POPULAR MUSIC

Anonymous Authors
Anonymous Affiliations
anonymous@ismir.net

ABSTRACT

Indian popular music, characterized by its unique scales and vocal styles, presents additional challenges for cover song identification. While there are culture-specific characteristics in tcovers of Indian popular songs, there are similar attributes that are culturally invariant. We leverage this insight by fine-tuning CQTNet, a state-of-the-art model pre-trained on a large dataset of Western cover songs. We notice an enhancement in results when fine-tuning is performed using the Indian popular songs dataset. To mitigate the problem of data scarcity, we conducted experiments involving data augmentation, creating modified cover versions with altered tempo and tonic. In our training, we investigated the effectiveness of Triplet Loss and its enhanced version, which takes supplementary positive and negative instances that exhibit varying degrees of similarity and dissimilarity. Considering the prominence of the vocal melody in Indian popular songs, we also investigated the effectiveness of separating the vocal from the audio and achieved significant improvement. Additionally, we investigated whether large audio models like HuBERT and MERT can provide any information to improve the results.

1. INTRODUCTION

Indian popular music, known for its lively melodies and rhythms, is hugely popular across India and many other countries in South Asia, with cover songs becoming a big part as artists rework old and new hits. Identifying these cover songs matters for improving music recommendations and protecting copyrights. Indian popular songs are inspired by the raga-based melodies of Indian classical music and the stylistic elements of Western music, blending these influences into a distinct musical identity. In the theory and practice of Western music, harmony is a core concept [1]. Unlike Western songs, Indian popular music gives more importance to the main vocal melody. The vocalists generally employ melismatic singing and ornamentations. Apart from these characteristics, most Indian popular songs exhibit a distinct melodic and rhythmic structure.

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The unique features and salient components of Indian popular music play a significant role in the similarity between original songs and their cover versions, as these elements tend to be retained in the covers. Given the importance of vocal melody in Indian popular music, cover versions typically retain the original melody with only minor changes. The design of many cover song identification methods has been motivated by the assumptions behind what makes a cover [2]. The prevailing notions of what constitutes a cover version in Indian popular music underpin this research inquiry and motivate the need for a separate investigation.

Earlier cover song identification approaches involved extracting features from audio signals and measuring the similarity between songs using these extracted features. Initial approaches relied on pitch detection and finding the optimal alignment between the query and the candidate song [3]. Quite a good number of approaches investigated the effectiveness of chroma features, which represent the energy distribution across different musical notes [4–6]. Dynamic programming-based approaches like Dynamic Time Warping (DTW) [7] and Smith-Waterman alignment [8, 9] enabled the sequence similarity.

Later, deep learning transformed cover song identification by automating feature extraction and improving accuracy, outperforming traditional methods. Some of these approaches aim to learn song representations that are subsequently used to calculate song similarities. Notable approaches in this direction include the siamese network-based method [10] and the CNN-based CQTNet [11]. Chang et al. [12] transformed cover song identification into an image classification task by applying convolutional neural networks to cross-similarity matrices derived from songs' chroma features [13]. ByteCover investigates a feature learning method with the ResNet model along with multi-loss training [14]. CoverHunter [15] uses a transformer model with attention and a combined focal, center, and triplet loss function.

There are no culture-specific approaches to cover song identification in any musical tradition. However, culture-specific investigations have been conducted in numerous other music information retrieval tasks. For instance, research has been conducted on pitch detection in Turkish music [16, 17], Indian popular and classical music [18, 19], Flamenco singing [20, 21], Arab music [22, 23]. The culture-specific attributes of music emotion detection have initiated investigations in Turkish music [24, 25], Indian

popular music [26, 27], Greek music [28].

Our research investigates cover song identification in Indian popular music, driven by two key questions:

1. Is a specialized model, trained exclusively on Indian popular music, necessary for effective cover song identification?
2. Do unique characteristics of Indian popular music necessitate custom-designed approaches for accurate cover song identification?

Being one of the best and widely experimented cover song identification methods, we choose CQTNet [11] as our baseline. Considering our limited dataset of Indian popular songs and the fact that there are cultural invariant similarity attributes between the original and the cover version of a song, we experiment with fine-tuning CQTNet with cover song dataset for Indian popular songs (CoversIndian). The performance gains with the fine-tuned model over the original CQTNet model provide a clear answer to the first research question. Leveraging the knowledge that vocal melody plays a dominant role in Indian popular songs, we investigate the effectiveness of isolating vocals from the audio signal. This pre-processing step yields improved results on the CoversIndian dataset, whereas it lowers the results on the Covers80 dataset [29]. This outcome answers our second research question, highlighting the importance of culturally informed approaches.

To address the challenge of limited training data, we explore a data augmentation technique. Additionally, we examine the impact of triplet loss [30] and propose a modification to the triplet loss tailored for CQTNet. While fine-tuning and vocal separation show significant improvements in results, certain combinations of the other proposed modifications also yield observable enhancements. To the best of our knowledge, this study is the first to specifically address cover song identification within the context of a particular music culture.

2. METHODOLOGY

2.1 CQTNet

CQTNet [11] has a CNN-based architecture that processes Constant-Q Transform (CQT) of the audio as input representations. The model effectively captures musical patterns across different pitch ranges through its unique architecture, featuring dilated convolutions and specialized filter designs in the initial layers. The model has been benchmarked on various datasets and attains a maximum MAP of 0.917 for the YouTube350 and 0.933 for the Mazurkas dataset. The model is primarily trained on large Western cover song datasets SHS100K. The model is trained for the label classification task, where the number of classes is the number of distinct original songs. The input CQT representations are passed through the convolution layers to fully connected Layers, and cross entropy loss is employed for label classification.

During inference, the embedding for a song is taken from the penultimate layer. The distance between the songs is computed by taking the cosine distance between the embeddings. The songs are ranked according to their cosine distance computed from the previous step. Considering the performance of CQTNet for cover song identification, we base our explorations on this model.

2.2 Fine-Tuning with Indian Cover Songs Dataset

Certain factors in cover song identification remain consistent across cultures. The original CQTNet model, trained on an extensive Western cover song dataset, has learned to identify the key factors that distinguish cover songs. Fine-tuning the model on the CoversIndian dataset enables targeted adaptation to Indian cover songs' cultural-musical attributes while maintaining transfer learning benefits from training on Western music. This is particularly important given the limited number of labeled data on Indian cover songs. In all our further experiments, we use the CQTNet model, which is fine-tuned with our CoversIndian dataset. Given the low number of classes to classify in the Indian songs dataset, we reduced the number of nodes in the final classification layer of the architecture.

2.3 Vocal Separation

Indian popular music is heavily influenced by Indian classical music and Western music. Even though the proportion of the blend is not generally uniform, the song's identity primarily lies in the predominant vocal melody. This suggests that the vocal component plays a more significant role in determining similarity. The importance of vocal melody motivated the experiments with extracting vocals and ignoring the accompanying instruments. Our experiments demonstrate that vocal separation is especially beneficial for Indian music, where vocal nuances are a key distinguishing factor.

2.4 Data Augmentation

The limited dataset and availability of only one cover song per original song pose challenges in learning diverse possible variations in a cover version of an original song. To compensate for this, we applied a basic data augmentation technique on the audio dataset. Feng et. al. [15] have experimented with data augmentation for cover song identification. We created 4 variants of a cover song with two pitch variations of +4 and -3 semitones, and two tempo variations of 1.2x and 0.8x were applied on the cover version. This results in a total of 5 cover versions for each original song in the training data. This would help the model to learn similarity with more variants.

2.5 Triplet Loss

When the cross-entropy loss helps in training the model to recognize similarities with a positive cover song, the triplet loss enables it to learn the dissimilarities with a non-cover song of the original song. Triplet loss minimizes the distance between anchor-positive pairs while maximizing the distance between anchor-negative pairs by a margin. The loss L is computed by taking the maximum between the

distance(Anchor-Positive, Anchor-Negative) and 0, ensuring that the loss remains non-negative.

$$L = \max\Big\{0,\, \alpha + \, d\big(A_{\text{ori}}, P_{\text{cover(i)}}\big) - d\big(A_{\text{ori}}, N_{\text{rand}}\big)\Big\} \tag{1}$$

where $d()$ is the distance function, A_{ori} represents the original song (anchor) embedding, $P_{\text{cover(i)}}$ denotes the i-th cover of the song, N_{rand} is a random negative example and α is the margin of separation between the positive and negative examples. Triplet loss has been experimented in cover song identification before in CoverHunter [15] and ByteCover [14].

2.6 Modified Triplet Loss Function

To further refine our embedding space, we modified the triplet Loss function to focus on the most challenging examples. It is very likely to have negative cover instances which are similar to the original song. The strong melody prominence in Indian popular songs increases the likelihood of negative cover instances closely resembling the positive cover instances of the original song. This modified loss encourages the network to learn more discriminative features by considering the subtle differences with the most similar negative examples. The modified triplet loss is defined as:

$$L = \max\Big\{0,\, \alpha + 2\, d\big(A_{\text{ori}}, P_{\text{cover(i)}}\big) - \Big[d\big(A_{\text{ori}}, N_{\text{sim}}\big) + d\big(A_{\text{ori}}, N_{\text{rand}}\big)\Big]\Big\} \tag{2}$$

where A_{ori} represents the original song (anchor) embedding, $P_{\text{cover(i)}}$ denotes the i-th cover of the song, N_{sim} is the most similar negative example, N_{rand} is a random negative example and α is the margin of separation between the positive and negative examples. We select the negative example exhibiting the smallest distance to the anchor along with a random negative example. We incorporate a weighting factor of 2 applied to the distance between anchor-positive instances. This balances the contributions of the positive and negative terms. As a result, the model achieves improved generalization across both general and edge-case negative examples.

The experiments with the modified triplet loss function involve two fine-tunings. The representation obtained from the model after the first fine-tuning is utilized to identify the most similar negative cover instance for each original song in the training data.

Algorithm2.6 describes the fine-tuning with the modified triplet loss. $\mathcal{M}$ is the model from the first fine-tuning. This model is used to compute the embeddings for the original and cover songs in the train data. A_{orig} represents the anchor embedding for each original song s and $Cover_A$ is one of the cover songs of A_{orig}; N_{rand} is a random negative example; and N_{sim} is the most similar negative example.

Algorithm 1 Modified Triplet Loss Fine-tuning

1: $\mathcal{M} \leftarrow$ Fine-tune on label classification.
2: Fine-tune embeddings using Modified Triplet Loss:
3: $E = get_embeddings(\mathcal{M}, D)$, where D is the dataset with instances as tuples (D_{orig}, D_{cover})
4: **for** each $A_{\text{orig}}, Cover_{\text{A}}$ in E **do**
5: $\quad C^A \leftarrow$ All covers of A_{orig}
6: $\quad N_{\text{rand}} :=$ random(E_{cover}), where $N_{rand} \notin C^A$
7: $\quad N_{\text{sim}} := \underset{E_{cover}(i) \notin C^A}{\text{argmin}} \; Dist(A_{\text{orig}}, E_{cover}(i))$,
8: $\quad L :=$ Triplet$\Big(A_{\text{orig}}, Cover_A, N_{\text{rand}} + N_{\text{sim}}\Big)$
9: **end for**
10: Update model parameters based on the computed loss.

2.7 Combining CQTNet and MERT/HuBERT Embeddings

Large audio representations are found helpful for various downstream tasks involving audio data. We investigate the potential of large audio models to gain insights into the relationship between a song and its cover version.

HuBERT [31] (Hidden unit BERT) is a state-of-the-art approach for self-supervised speech representation learning in tasks like speech recognition, generation, compression, and speaker verification in both clean and noisy conditions. MERT [32](Music undERstanding model with large-scale self-supervised Training) adapts this framework specifically for music audio. It tries to address the unique challenges of representing the tonal and pitched characteristics of musical content. MERT has demonstrated benchmarking performance across different Music Information Retrieval tasks like beat tracking, singer identification, music tagging, key detection, genre classification, and more. Both models leverage self-supervised learning to create rich audio representations without requiring labeled data. This makes them powerful foundations for various downstream audio analysis applications.

We experimented with an approach that concatenates CQT features with MERT embeddings. We aimed to determine whether the complementary information captured by MERT/HuBERT can enhance cover song similarity when combined with the traditional CQT features. The MERT embeddings are computed using the pre-trained model (without updating during training) and are passed through feed-forward layers to capture only the relevant features. We obtain the embeddings from CQTNet by processing the CQT features of the same audio sample. The embeddings from the MERT feed-forward layer and the CQTNet model are combined and then passed through a final feed-forward layer. Figure 1 describes the architecture of this approach. Cosine similarity is employed during the inference.

3. EXPERIMENTS

3.1 Dataset

The CoversIndian dataset comprises audio data from original songs and their covers. The dataset contains 245 song pairs from 2 different Indian languages: Hindi and Malayalam. Of these, 126 song pairs are for fine-tuning, and 119 song pairs are for Testing.

We have also used Covers80 dataset [29] as test set for showing certain results on a Western music dataset. It

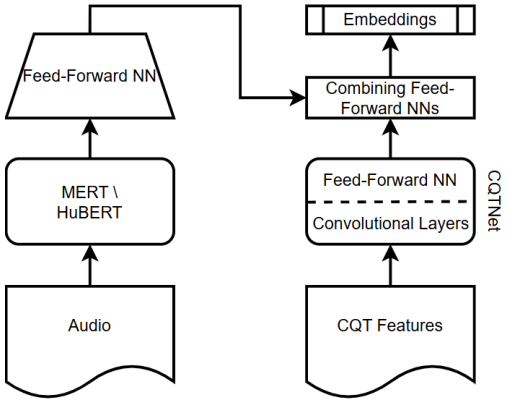


Figure 1. CQTNet+MERT/HuBERT Architecture

contains 80 pairs of Western songs.

3.2 Experimental Setup

We computed the CQT representations using the Librosa library [33] for the audio datasets to train and evaluate the models. Two sets of data preprocessing were applied on the CoversIndian and WesternCovers datasets, as mentioned in the Methodology sections 2.3 and Section 2.4.

Vocal Separation from the datasets were done using the API from music.ai (erstwhile Moises) [34]. We compute CQT representations of the vocal-separated audio to train and evaluate the models.

Data Augmentation techniques were applied to the cover versions of both the audio and vocal-separated audio datasets. Data augmentation can result in generating 4 additional versions for each cover song, varying in pitch and tempo. The data augmentations of pitch shift and tempo variations, as discussed in Section 2.4, were applied using the Librosa library. We compute the CQT representations for all the augmented data and use them for training and analysing the models.

Experiments were carried out on Intel Gaudi3 AI accelerators (128GB) and an NVIDIA A100 GPU (80GB). For reproducibility and consistency across all experiments, we conducted the experiments in Random Seed(0) environment. All label classifications were fine-tuned for 200 epochs, and all Triplet Loss fine-tunings for 150 epochs. A batch size of 60 was used with shuffle set to False; This helps to process 10 sets of 6 song versions, 1 original and 5 augmented covers for Triplet Loss fine-tuning.

3.3 Evaluation

We used Mean Average Precision (MAP), Precision@10 (Top10), and mean rank of the first correctly identified cover (Rank1) as our evaluation metrics. All the metrics are widely used for Cover Song identification problems.

4. RESULTS

The following acronyms have been used in the results tables.

Acronym	Details
VocalSep	Vocal Separated Audio
DA	Data Augmentation
Trip.Loss	Triplet Loss
Mod Trip.Loss	Modified Triplet Loss

Table 1. Acronym Reference

4.1 Baseline CQTNet Vs Fine-tuned CQTNet

This experiment fine-tunes the original CQTNet model pre-trained exclusively on Western cover songs to the CoversIndian dataset.

Model	MAP	Top10	Rank1
Baseline CQTNet	0.7238	0.0840	8.87
Fine-tuned CQTNet	0.7379	0.0819	11.53

Table 2. Results of baseline CQTNet Vs CQTNet fine-tuned with CoversIndian

4.2 Vocal Separation

In this experiment, we try to see the impact of vocal separation on CoversIndian and Covers80 test set. As reported by Zhesong et. al. [11], the baseline CQTNet gives high MAP on Western covers testset Covers80 with source audio. The baseline CQTNet does not give similar results on CoversIndian test set. Results are shown in Table 3. As hypothesized, vocal separation yielded a notable improvement for CoversIndian. We can see this improvement on CoversIndian test set in both the experiments using baseline CQTNet and fine-tuned CQTNet. However, there is a degradation in MAP for Covers80. The results strongly back the claim that Indian songs are predominantly based on vocal melody. However, it is not significant to Western music.

Dataset	Method	MAP	Top10	Rank1
CoversIndian	Baseline CQTNet	0.7238	0.0840	8.87
	Baseline CQTNet + VS	0.8629	0.0937	3.95
	Fine-tuned CQTNet	0.7379	0.0819	11.53
	Fine-tuned CQTNet + VS	0.9001	0.0962	2.51
Covers80	Baseline CQTNet	0.8768	0.0944	3.30
	Baseline CQTNet + VS	0.7181	0.0819	8.79

Table 3. Results with vocal separation

The vocal separated pre-processed audio consistently led to better performance compared to source audio inputs. Notably, similar experiments conducted on Western cover songs demonstrated that vocal separation led to lower MAP values compared to their original cover songs, underscoring that Indian cover songs inherently emphasize vocal nuances- which in turn contribute to the improved MAP observed.

4.3 Data Augmentation

CQTNet was fine-tuned with the Augmented CoversIndian dataset. Table 5 shows the results with data augmentation.

Method	MAP	Top10	Rank1
VocalSep	0.9001	0.0962	2.51
DA+VocalSep	0.9131	0.0958	2.91
Trip.Loss+VocalSep	0.8839	0.0979	3.01
Trip.Loss+DA+VocalSep	0.8953	0.0971	3.33
Mod Trip.Loss+VocalSep	0.8922	0.0962	2.14
Mod Trip.Loss+DA+VocalSep	**0.9234**	0.0975	2.27

Table 4. Consolidated results on CoversIndian test set. Mentioned modifications applied on fine-tuned model with CoversIndian train set. (Refer Table 4 for acronyms used).

The results can be compared againt Table 3. We can see that the data augmentation gives improvement on vocal separated data, but gives no improvement on source audio data.

Method	MAP	Top10	Rank1
DA + Source Audio	0.7307	0.0857	9.89
DA + VocalSep Audio	0.9131	0.0958	2.91

Table 5. Results after applying data augmentation on CoversIndian testset

4.4 Triplet Loss Function

We used the Triplet Loss function to analyse how well it captures the variations in cover song versions of source and vocal-separated audio. In this experiment, we use the Triplet Loss function for the penultimate embedding layer instead of label classification using Cross Entropy at the last layer. The results in Table 6 show that enabling triplet loss gives a significant improvement on the source audio data with data augmentation. However, on data augmented vocal separated triplet loss leads to performance degradation

Method	MAP	Top10	Rank1
DA + Source Audio	0.7696	0.0861	6.97
DA + VocalSep Audio	0.9079	0.0975	2.56

Table 6. Results with triplet loss on CoversIndian test set

4.5 Modified Triplet Loss

As mentioned in the 2.6, this involves 2 levels of fine-tuning to get the most similar negative cover instance for each original song in the train set.

Method	MAP	Top10	Rank1
DA + Source Audio	0.7901	0.0899	5.83
DA + VocalSep Audio	0.9234	0.0975	2.27

Table 7. Results with modified triplet loss on CoversIndian test set.

The Modified Triplet Loss function showed promising improvements on the Indian cover song datasets. The result in Table 7 suggests that the Modified Triplet loss helps the embeddings to capture the subtle differences with the negative cover instances. Our comparative evaluation demonstrates that the Modified Triplet Loss outperforms Triplet Loss in capturing variations in augmented cover songs on source audio and vocal separated data.

4.6 Additional Evaluation of HuBERT and MERT

Table 8 summarizes our experiments with MERT-330M and the HuBERT-large model. We made the MERT/HuBERT weights non-trainable during the experiment and passed the pre-trained representations to CQTNet as detailed in Section 2.7. No improvements were seen in these experiments, so we did not explore this direction further. However, merging embeddings based on attention can be tried in future.

Model on CoversIndian	MAP	Top10	Rank1
CQTNet+MERT	0.5017	0.0664	28.03
CQTNet+Hubert	0.4466	0.0601	37.60

Table 8. Results with architecture combining MERT/HuBERT with CQTNet

5. CONCLUSION

This study successfully adapts a state-of-the-art cover song identification method CQTNet, to the unique features of Indian popular music, addressing challenges posed by its distinct melodic and vocal characteristics. By fine-tuning CQTNet on the CoversIndian dataset, we demonstrate significant performance improvements, particularly when leveraging vocal separation techniques that emphasize the prominence of vocal melodies in Indian music. Data augmentation further mitigates the issue of limited training data, enabling the model to learn from diverse variations of cover songs.

The introduction of a modified triplet loss function proves effective in capturing subtle differences between similar negative cover instances and positive covers, enhancing the model's ability to discriminate between challenging examples. While large audio models like HuBERT and MERT did not yield improvements in this context, future exploration may unlock their potential.

Our findings highlight the importance of culturally informed pre-processing steps and loss functions tailored to specific musical traditions. This research lays the groundwork for further investigations into adapting cover song identification to diverse musical genres and traditions.

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